

# Review of Trigonometric functions

What are trig Functions?

- they relate angle to ratios in a right triangle  
and the unit circle.

The way 3:

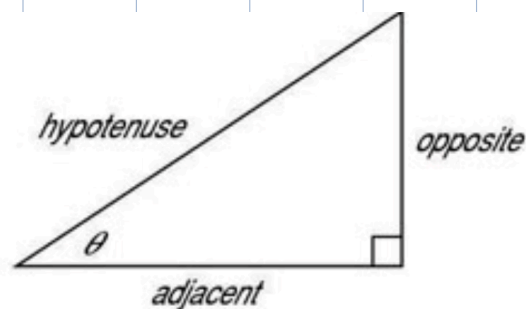
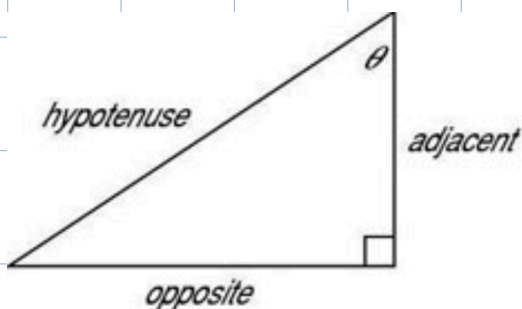
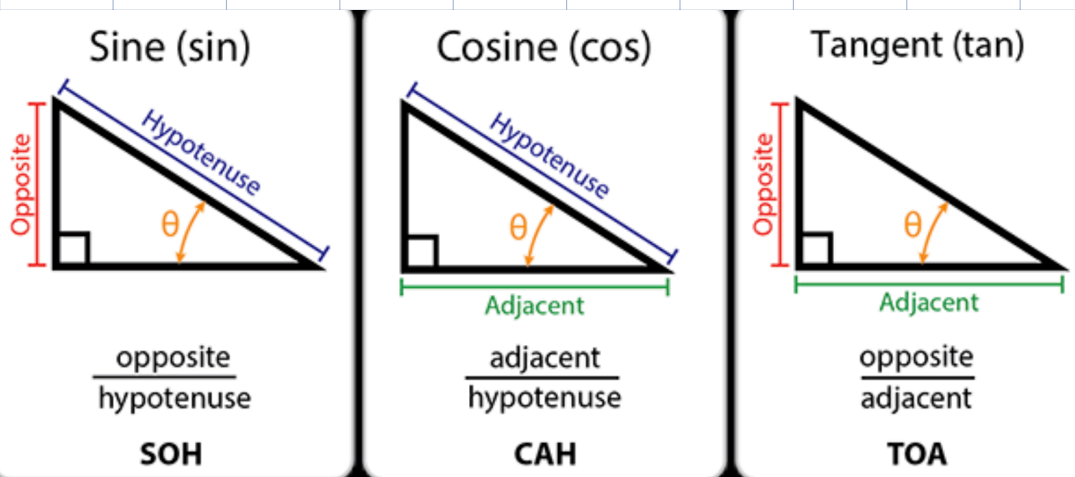
$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$$

Finding  $\theta$  of a triangle:

$$\theta = \tan^{-1} \left( \frac{\text{opposite}}{\text{adjacent}} \right)$$

$$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$$



Example:

Opposite = 3

$$\sin \theta = \frac{3}{5}, \quad \cos \theta = \frac{4}{5}, \quad \tan \theta = \frac{3}{4}$$

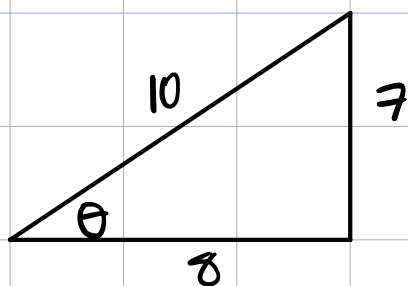
Adjacent = 4

hypotenuse = 5

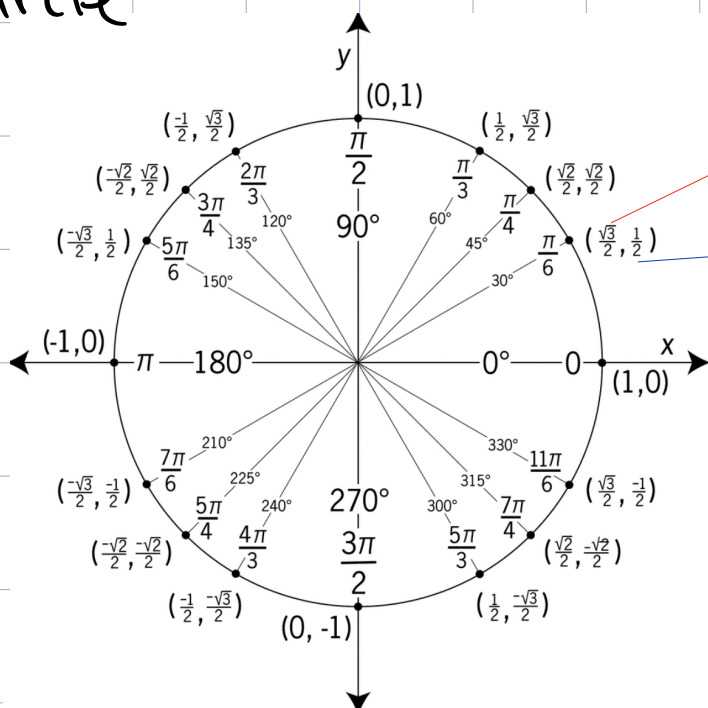
Example

Find  $\sin(\theta)$ ,  $\cos(\theta)$ ,  $\tan(\theta)$ , and  $\theta$

given!



 Unit Circle



Cosine = x-Coordinate

Sine = y-Coordinate